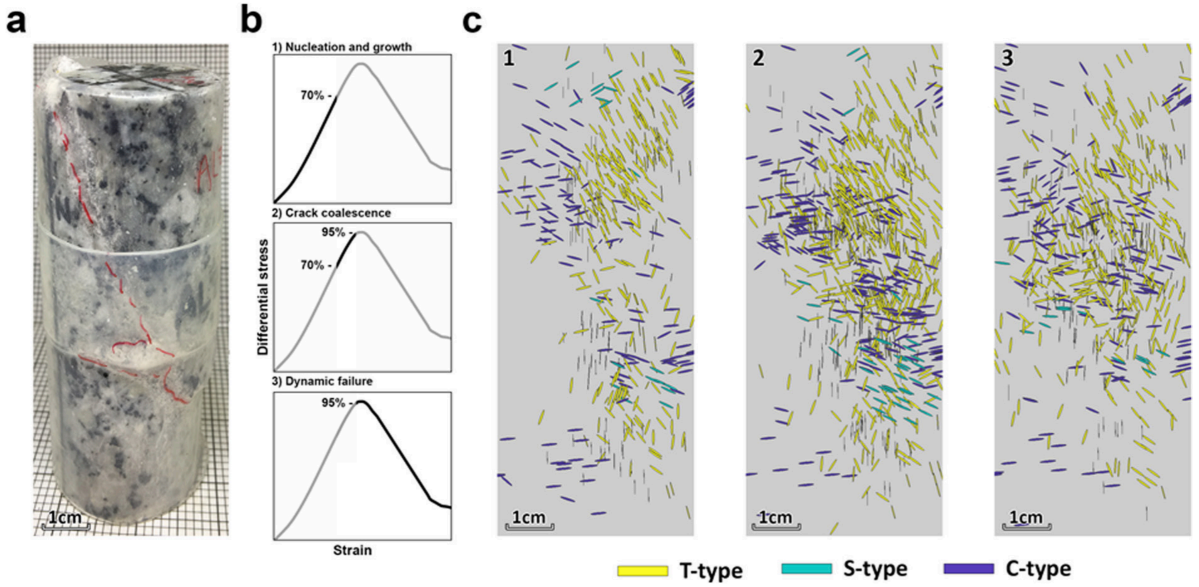
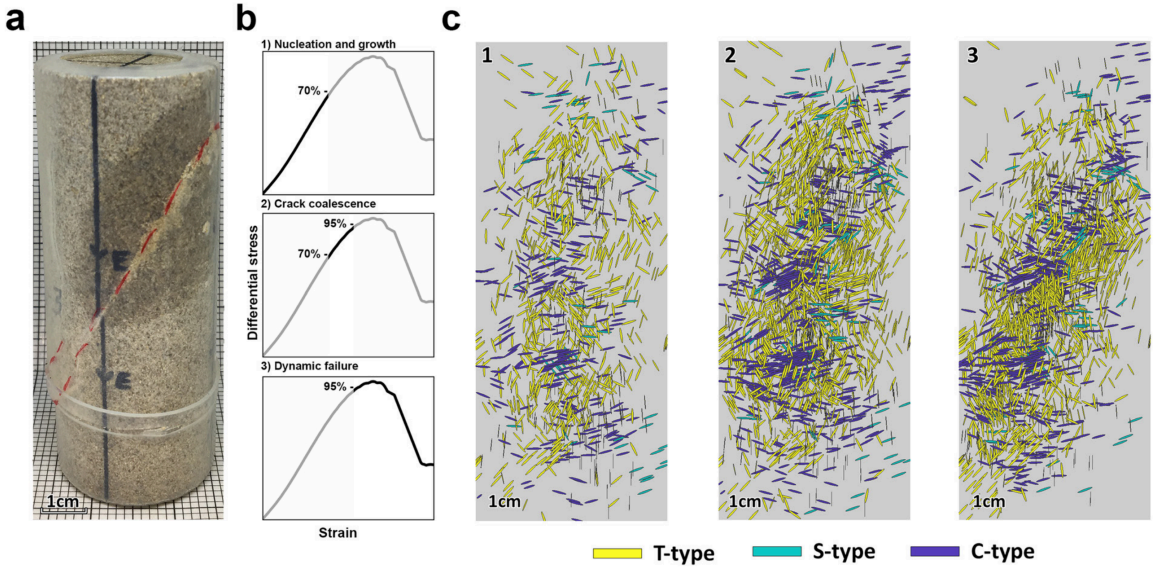
Granite vs sandstone: where do different cracks form?

Identify the tensile (yellow), shear (cyan) and compactive (purple) sources.



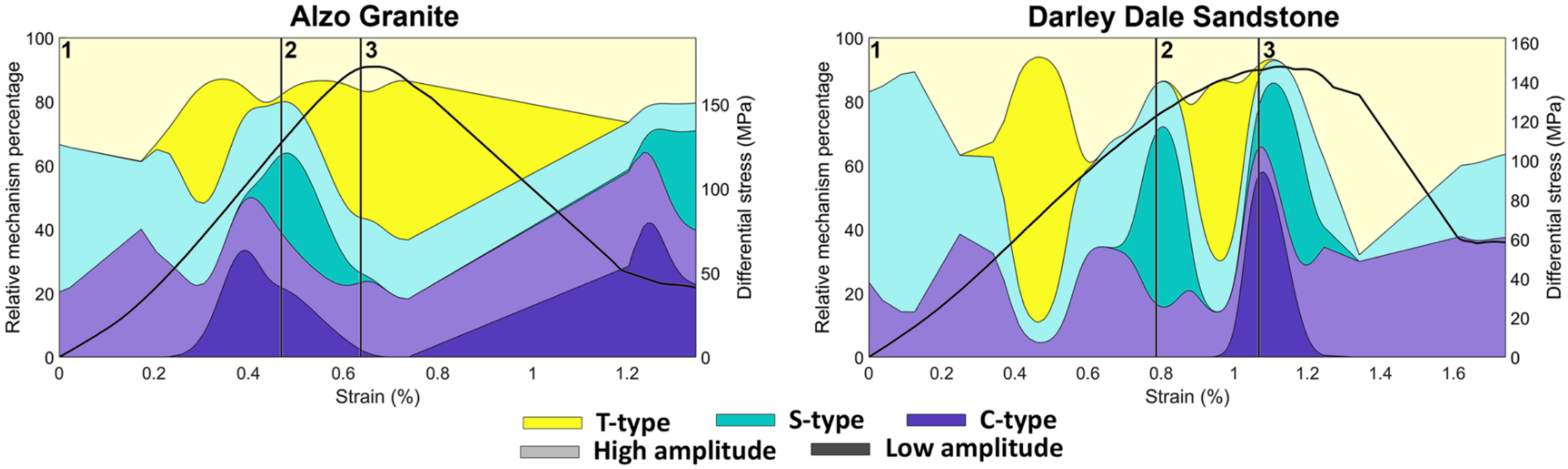


AE source mechanisms evolve in space through nucleation, coalescence and failure.

King et al. (2021), JGR Solid Earth, Figures 5–6 · DOI: 10.1029/2020JB021059

The crack population changes during loading

Compare the proportions of opening, shear and compactive events in the two rocks.

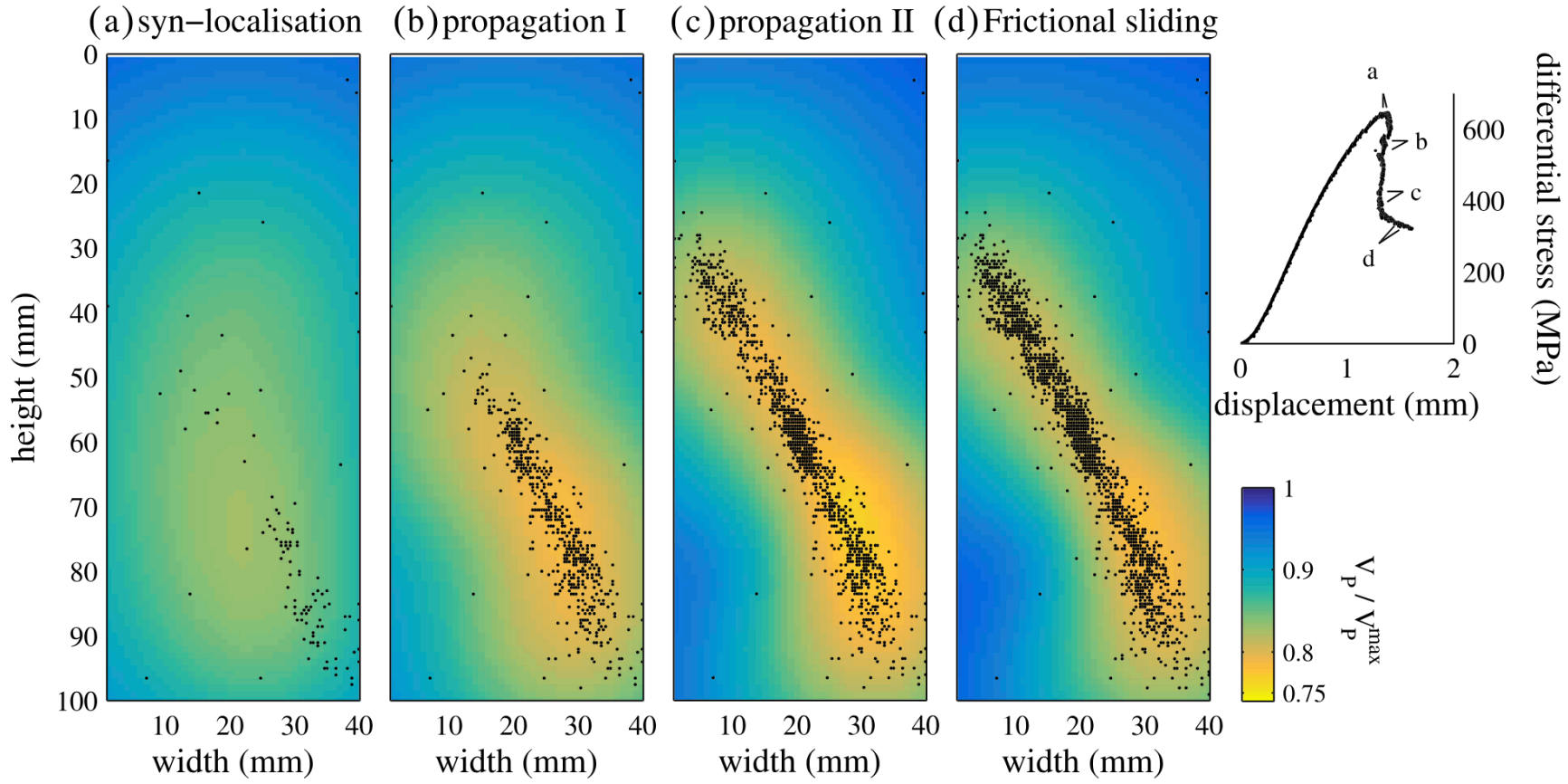


Opening and compaction coexist; their proportions vary with lithology and strain.

King et al. (2021), Figure 9. T = tensile; S = shear; C = compactive.

AE locations + P-wave tomography track a propagating fault

Where do the AE hypocenters cluster relative to the developing low-velocity zone?



A propagating AE cluster accompanies a growing low-V_p damage zone.

Aben et al. (2019), Geophys. Res. Lett., Figure 2 · DOI: 10.1029/2019GL083040

COMPLETE STORY

From loading to a fault: the acoustic chain of inference

LOAD

stress concentrates

→

MICROCRACK

rapid strain radiates

→

AE CATALOGUE

detect · locate · size

→

LOCALIZE

patch grows + coalesces

→

FAULT

VP deficit + crack fabric

OBSERVATION	voltage · arrival time · stress · travel time · micrograph etc.
INFERENCE	event · location · mechanism · damage zone · nucleation

Good interpretation never hides the model and assumptions between the two.

No single observable is the fault — together they constrain a process

AE

rapid increments

when + where damage is active

VP

elastic consequence

where crack compliance accumulates

MICROSTRUCTURE

physical traces

what cracks actually formed

Convergence strengthens interpretation; disagreement is scientifically useful.

Best claim: a localized, growing damage zone precedes and becomes a fault.

LEARNING CHECK

Can you now listen to a forming fault?

1. **Why can one AE event produce many recorded hits?**

2. **How can an evolving VP field bias AE locations?**

3. **What is the clearest evidence that damage has localized?**

4. **Why are AE, VP and microscopy complementary rather than interchangeable?**

One-sentence exit claim: Before a fault fails dynamically, _____.

Next: from fault formation to frictional sliding and stick–slip

Next: measure acoustic activity during a UCS test

Before loading

Check sample, sensors, channels and synchronization.

During loading

Follow force–shortening and AE signals. Document threshold and noise.

After acquisition

Extract detections, build a catalogue and compute AE rate.



The central question: when does AE activity change, and what does that tell us about fracture?